Notifications – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System
> **Parent Module:** Communication · **Child Module:** 05.1 Notifications (incl. deadline reminders engine)
> **Version:** 1.0 · **Last Updated:** 31-Oct-2026
> **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
> **Author:** Jojo George (jojogeorge3344@gmail.com)
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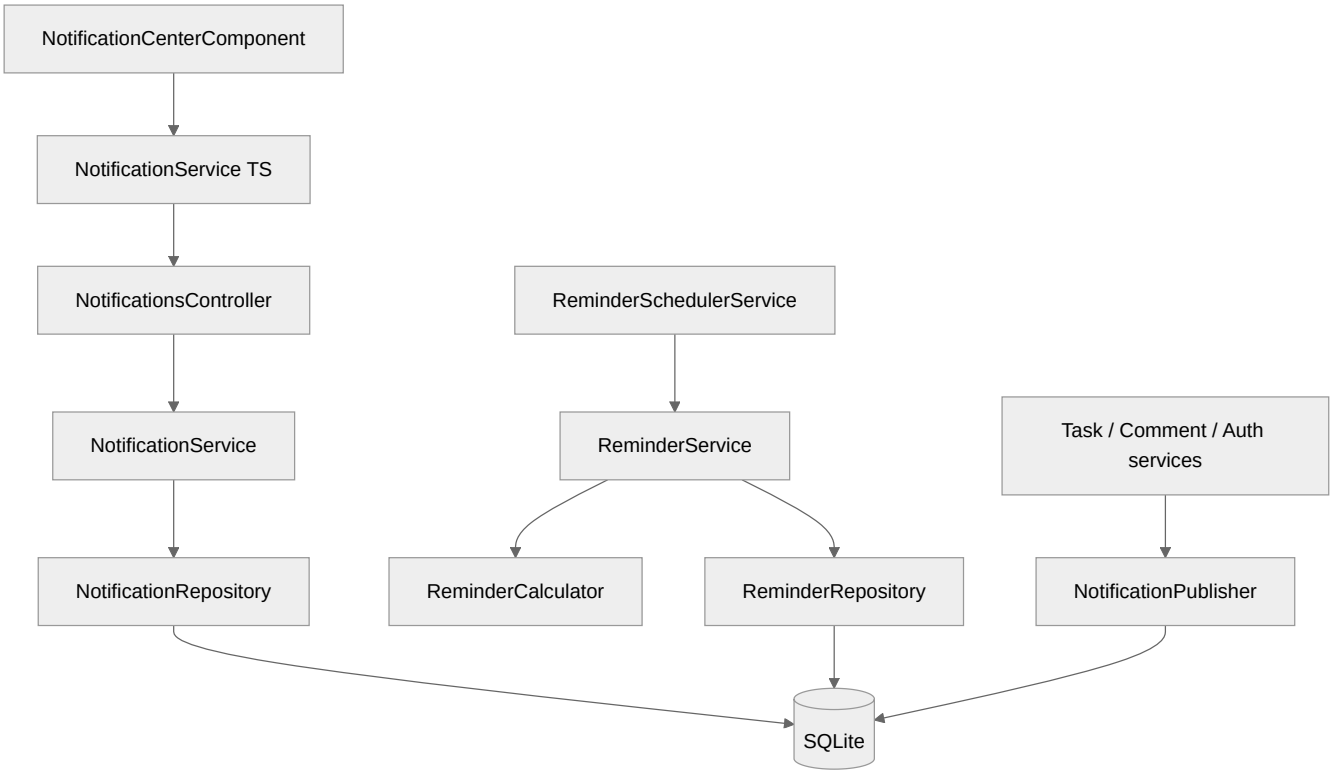
1. Technical Overview

The Notifications child module has three parts:

- **Notifications (read side)**: `NotificationsController` → `NotificationService` → `NotificationRepository`. It serves the caller's own rows from the `Notifications` table for the Notification Center (BRD Screen 12).
- **Notification publishing (write side)**: `INotificationPublisher` / `NotificationPublisher` is an append-only seam that other modules call (Monitoring, Auth). It skips writes while the `Notification.InAppEnabled` application setting is `false`.
- **Deadline reminder engine**: `ReminderSchedulerService` is a `BackgroundService` running a `PeriodicTimer`. On each tick it calls `ReminderService.GenerateDueAsync()`. That method uses the pure `ReminderCalculator` to decide which reminders are due today. It then inserts into `ReminderHistory` with an insert-or-ignore on a unique index, and only when a row was added does it write a `Notification` and, optionally, a system `MailMessage` (Mail child module tables).

Every endpoint is under `/api/v1/notifications`, requires a JWT (`[Authorize]`) and a permission policy `perm:<key>`, and resolves the caller from the `ClaimTypes.NameIdentifier` claim.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Communication/`	`NotificationsController.cs`
Service – Notifications	`TaskFlow.Api/Services/Communication/Notification/Notifications/`	`INotificationService.cs`, `NotificationService.cs`
Service – Publishing	`TaskFlow.Api/Services/Communication/Notification/Publishing/`	`INotificationPublisher.cs`, `NotificationPublisher.cs`
Service – Reminders	`TaskFlow.Api/Services/Communication/Notification/Reminders/`	`IReminderService.cs`, `ReminderService.cs`, `ReminderSchedulerService.cs`
Repository – Notifications	`TaskFlow.Api/Repositories/Communication/Notification/Notifications/`	`INotificationRepository.cs`, `NotificationRepository.cs`
Repository – Reminders	`TaskFlow.Api/Repositories/Communication/Notification/Reminders/`	`IReminderRepository.cs`, `ReminderRepository.cs`, `ReminderCandidate.cs`, `DeliveredReminder.cs`, `TaskRecipientLink.cs`
DTOs	`TaskFlow.Api/Dtos/Communication/Notification/`	`NotificationDtos.cs` (`NotificationDto`, `NotificationFilter`, `UnreadCountDto`)

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Layer	Path	Files
Rules	`TaskFlow.Api/Common/Rules/`	`ReminderCalculator.cs` (`ReminderTaskSnapshot`, `DueReminder`)
Entities	`TaskFlow.Api/Models/Entities/`	`Notification.cs`, `ReminderHistory.cs`
Enums	`TaskFlow.Api/Models/Enums/`	`NotificationType.cs`, `ReminderType.cs`, `ReminderStatus.cs`, `ReminderFrequency.cs`
Config	`TaskFlow.Api/Config/`	`ReminderEngineOptions.cs`, `MailOptions.cs` (`AppBaseUrl` used in reminder mails)
Clock (Settings module)	`TaskFlow.Api/Services/Settings/ApplicationSettings/Clock/`	`IAppClock.cs`, `AppClock.cs` (application timezone)
Persistence	`TaskFlow.Api/Persistence/`	migrations `20261013075057_AddTaskTracking` (creates `Notifications`), `20261017062735_AddReminderEngine` (creates `ReminderHistory`)

4. Frontend File Structure

Type	Path	Purpose
Route	`src/app/modules/communication/communication.routes.ts`	`/notifications` (requires `notifications.view`), lazy-loaded with `permissionGuard`
Page component	`src/app/modules/communication/notification/pages/notification-center.component.ts/.html`	Notification Center: filters, list, paging, mark read / mark all
Page model	`src/app/modules/communication/notification/pages/notification-center.model.ts`	`ReadState` type
API service	`src/app/modules/communication/notification/services/notification.service.ts`	HTTP calls plus the shared `unreadCount` signal used by the header bell
Models	`src/app/modules/communication/notification/models/notification.model.ts`	`NotificationType`, `NOTIFICATION_TYPES`, `NotificationItem`, `NotificationFilter`, `UnreadCount`
Permission constants	`src/app/shared/auth/permissions.ts`	`NotificationsView`, `NotificationsManage`
Menu	`src/app/layouts/nav-items.ts`	"Communication Management" → Notifications (`notifications.view`)
Shell	`src/app/layouts/shell.component.ts/.html`	Header bell (`canSeeNotifications`) with unread dot. The count refreshes on navigation and on a 15 s interval (`SESSION_CHECK_INTERVAL_MS`).

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5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/notifications`	Caller's notifications, newest first, filter by `type`, `isRead`; paged (`page`, `pageSize`)	`notifications.view`	`NotificationFilter` (query)	`PagedResult<NotificationDto>`
2	GET	`/api/v1/notifications/unread-count`	Unread count for the header badge	`notifications.view`	–	`UnreadCountDto`
3	POST	`/api/v1/notifications/{id:int}/read`	Mark one owned notification read (idempotent)	`notifications.manage`	–	`204 No Content` / `404`
4	POST	`/api/v1/notifications/read-all`	Mark all of the caller's notifications read	`notifications.manage`	–	`204 No Content`

No endpoint exists for the reminder engine. It runs in-process (see 5.3).

5.2 Request / Response Samples

```
// GET /api/v1/notifications?type=Overdue&isRead=false&page=1&pageSize=20
{
  "items": [
    {
      "id": 51,
      "type": "Overdue",
      "message": "Overdue: \"Prepare Q3 report\" was due on 29 Oct 2026, 17:00.",
      "isRead": false,
      "createdAt": "2026-10-31T00:01:00Z",
      "taskId": 12,
      "taskCode": "TSK-0012"
    }
  ],
  "total": 1,
  "page": 1,
  "pageSize": 20
}
```

```
// GET /api/v1/notifications/unread-count  ->  200 OK
{ "count": 3 }
```

`PagedResult<T>` is `record PagedResult<T>(Items, Total, Page, PageSize)` in `Common/Models/PagedResult.cs`. The id, title and task-code values in the sample are illustrative.

5.3 Service Layer Logic

`NotificationService.ListAsync(int userId, NotificationFilter filter, CancellationToken)`

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1. Clamps `Page` to at least 1. If `PageSize` is below 1 or above 100, it uses 20.
2. Calls `INotificationRepository.ListAsync(userId, filter.Type, filter.IsRead, page, pageSize)`.

`GetUnreadCountAsync(userId)` calls `CountUnreadAsync`.

`MarkReadAsync(userId, notificationId)`

1. Calls `GetOwnForUpdateAsync`, which matches on Id and UserId. It returns `false` (404) when the row is not found.
2. If the row is unread, it sets `IsRead = true` and saves. It returns `true`, so repeat calls are idempotent.

`MarkAllReadAsync(userId)` loads `GetUnreadForUpdateAsync`, sets `IsRead = true` on each row and saves once. When there are no unread rows it does nothing.

`NotificationPublisher.PublishAsync(int userId, int? taskId, NotificationType type, string message)`

1. Reads the application setting `Notification.InAppEnabled`. If it is missing or cannot be parsed, it is treated as enabled.
2. If the setting is disabled, it returns without writing anything.
3. Otherwise it adds a `Notification` (`IsRead = false`, `CreatedAt = TimeProvider` UTC now) and calls `SaveChangesAsync`.

Callers of the publisher (side effects from other modules):

Caller	Type	Message
`TaskService` (create, inside the transaction)	`TaskAssigned`	`A task was assigned to you: {title}`
`TaskService` (reassign)	`Reassigned`	`A task was reassigned to you: {title}. Reason: {reason}` for the new assignee, otherwise `{actor} reassigned task "{title}" ({code}) from {old} to {new}. Reason: {reason}`
`TaskService` (worked hours)	`System`	`{actor} added/updated/cleared … on task {title}. Total worked now: …`
`TaskCommentService`	`TaskComment`	`{who} {verb} a comment on "{title}": {detail}`
`MfaService` (admin reset)	`System`	`An administrator reset your authenticator MFA. …`
`PasswordRecoveryService`	`System`	`Your password was reset with your authenticator app. …`

Reminder engine: `ReminderSchedulerService.ExecuteAsync`

1. If `Reminders:SchedulerEnabled` is false, it logs "Reminder scheduler is disabled by configuration." and exits.
2. Runs one pass straight away, then one per `PeriodicTimer` tick of `Max(1, Reminders:TickSeconds)` seconds.
3. `RunPassAsync` creates a DI scope and resolves `IReminderService`. Any exception is logged as "Reminder pass failed." and swallowed, so the next tick retries.

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`ReminderService.GenerateDueAsync()`

1. Reads the settings from `IReminderSettingsService.GetAsync()`. If `Enabled` is false, it returns 0.
2. Gets the timezone from `IAppClock.GetTimeZoneAsync()` (the `Application.Timezone` setting, falling back to `Application:Timezone` in config, then UTC) and the current UTC time.
3. Calls `IReminderRepository.GetOpenTasksDueByAsync(now + 32 days)`. This returns tasks that are not deleted, not Completed or Cancelled, and have a deadline no later than that date. There is no lower bound, so overdue tasks are included.
4. For each task, calls `ReminderCalculator.DueReminders(snapshot, nowUtc, tz, settings.DeadlineDayEnabled)`, which returns zero or one `DueReminder` for **today**:
   - `daysUntilDeadline > 0`: `SecondReminder` if it equals `SecondReminderDays` (when that is above 0), otherwise `FirstReminder` if it equals `FirstReminderDays`. Equal intervals collapse to one reminder.
   - `daysUntilDeadline = 0`: `DueToday` if `now < deadline` and either the global deadline-day setting is on or the task's first or second interval is 0.
   - `daysUntilDeadline < 0`: `Overdue` when `daysOverdue % step == 0`, where the step is 1 for Daily and 7 for Weekly.
   - `ScheduledForUtc` = local midnight of today, converted to UTC. The calculation handles DST gaps and ambiguous times.
5. `LoadRecipientsAsync` takes `TaskRecipients.UserId` plus `Task.CreatedByUserId`, removes duplicates, and keeps only users returned by `IEmployeeRepository.GetActiveByIdsAsync`.
6. `LoadDeliveredAsync` pre-reads `ReminderHistory` rows for today's slots, so that already-delivered reminders are skipped without taking the SQLite write lock.
7. Builds the message with `BuildMessage`. If `settings.MailEnabled` is on, it also builds a mail with `BuildMail`: the subject is `Overdue task: `, `Task due today: ` or `Task reminder: ` plus the title, truncated to 200 characters. The body is the message, then `Open the task: {AppBaseUrl}/tasks/{id}` when `Mail:AppBaseUrl` is set, then "This is an automatic reminder from TaskFlow."
8. `TryDeliverAsync` runs one transaction per recipient and reminder (`IUnitOfWork.ExecuteInTransactionAsync`):
   - `ReminderRepository.TryRecordAsync` runs a raw `INSERT … SELECT … WHERE EXISTS (task still live, open, same Deadline) ON CONFLICT (TaskId, RecipientUserId, ReminderType, ScheduledFor) DO NOTHING`. When it inserts no row, nothing else is written.
   - It adds a `Notification` directly through the repository. The type is `Overdue` or `DeadlineReminder`.
   - When mail is enabled, it adds a `MailMessage` (`IsSystem = true`, `SenderUserId = null`, `TaskId`, and `ReminderHistoryId` from `GetIdAsync`), plus a `MailRecipient` (To) and a `MailboxItem` (Inbox) for the recipient.
   - It saves and commits, then calls `DetachAll()`.
9. Logs "Generated {Count} deadline reminder notification(s)." and returns the number of rows written.

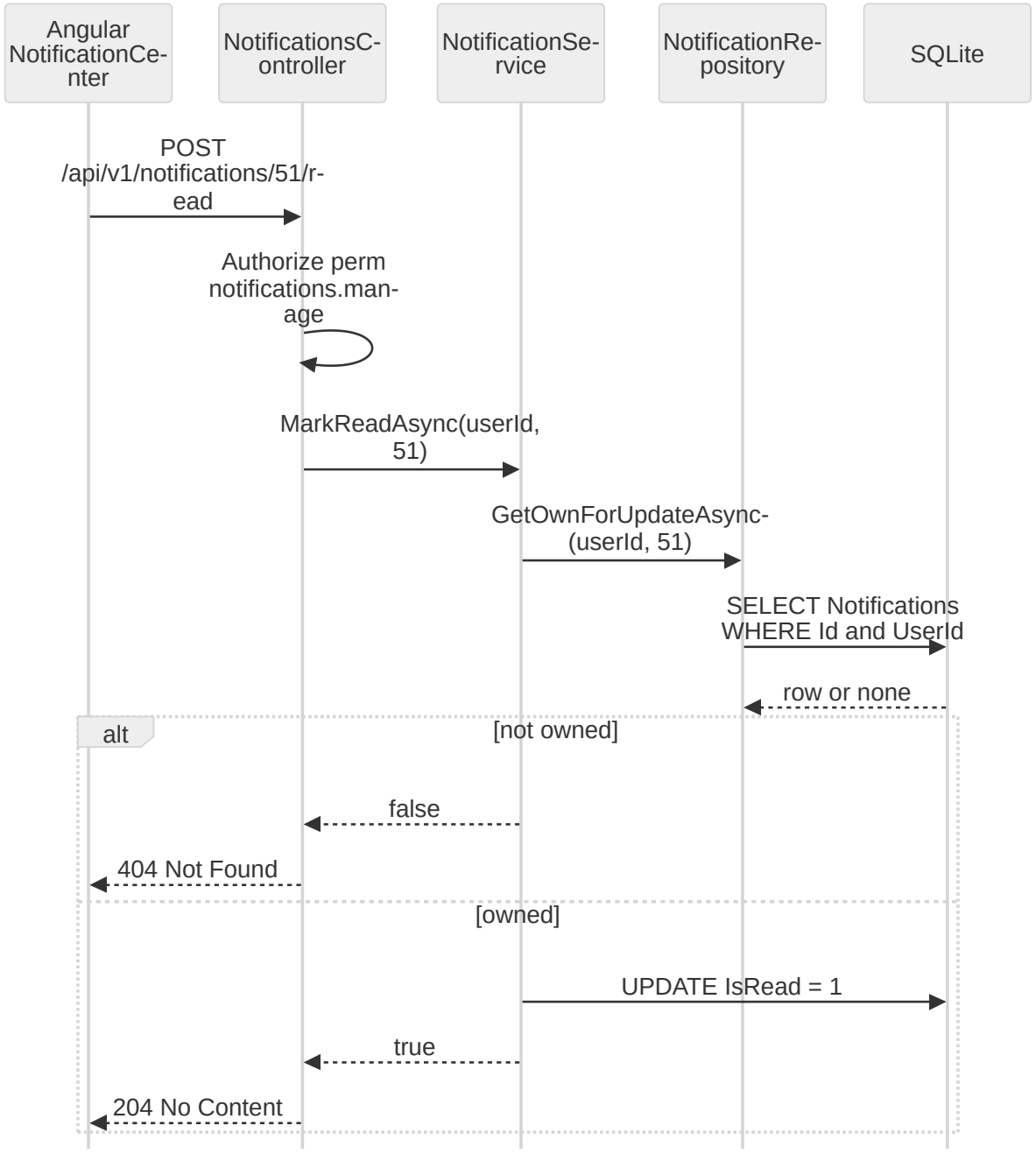
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5.4 Repository Layer

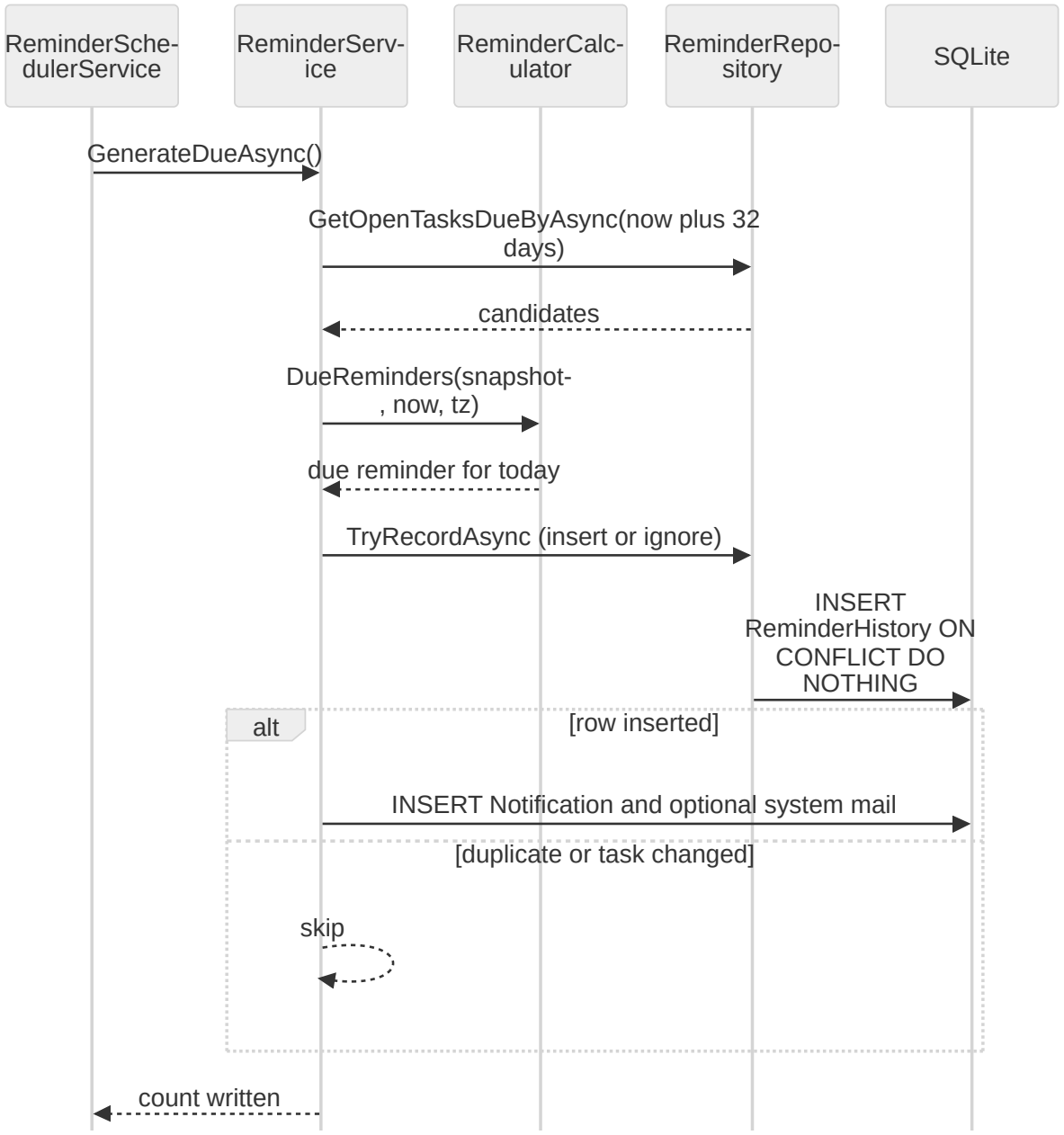
Method	Description	Query Notes
`NotificationRepository.ListAsync(userId, type, isRead, page, pageSize)`	One page of the user's notifications	`AsNoTracking`, `Where UserId`, optional `Type` and `IsRead`, `Include(Task)`, order by `CreatedAt` desc then `Id` desc. `ToDto` drops `TaskId` / `TaskCode` when the task is missing or `IsDeleted`.
`CountUnreadAsync(userId)`	Unread count	`CountAsync(UserId && !IsRead)`, served by index `(UserId, IsRead)`
`GetOwnForUpdateAsync(userId, id)`	Tracked own row	`FirstOrDefault(Id && UserId)`
`GetUnreadForUpdateAsync(userId)`	Tracked unread rows	`Where(UserId && !IsRead)`
`ReminderRepository.GetOpenTasksDueByAsync(dueBy)`	Reminder candidates	`Db.Tasks`, `AsNoTracking`, projects into `ReminderCandidate` and `ReminderTaskSnapshot`
`GetDeliveredAsync(taskIds, earliestSlot)`	Already-recorded reminders	`ScheduledFor >= earliestSlot`
`GetTaskRecipientsAsync(taskIds)`	Task notification recipients	`Db.TaskRecipients` projected into `TaskRecipientLink`
`TryRecordAsync(...)`	Idempotent history insert	Raw SQL `ExecuteSqlInterpolatedAsync`. Writes `Status = 'Sent'`, `ProcessedAt = now`, `DeliveryAttempts = 0`.
`GetIdAsync(...)`	Id of the history row just inserted	`SingleAsync` on the unique key

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5.5 Sequence Diagrams



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6. Database Design

Column types are given as C# type → SQLite storage. Enums that use `HasConversion<string>()` are stored as TEXT.

Table	Column	Type	Constraints	Description
`Notifications`	`Id`	int → INTEGER	PK, autoincrement	–
`Notifications`	`UserId`	int → INTEGER	FK → `Users.Id`, Not Null	Recipient
`Notifications`	`TaskId`	int? → INTEGER	FK → `Tasks.Id`, nullable	Related task

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Table	Column	Type	Constraints	Description
`Notifications`	`Type`	`NotificationType` → TEXT(32)	Not Null	TaskAssigned, Reassigned, DeadlineReminder, Overdue, System, TaskComment
`Notifications`	`Message`	string → TEXT	Required	Display text
`Notifications`	`IsRead`	bool → INTEGER	Not Null	–
`Notifications`	`CreatedAt`	DateTime → TEXT	Not Null	UTC
`Notifications`	–	–	Index `(UserId, IsRead)`	Unread count / filter
`ReminderHistory`	`Id`	int → INTEGER	PK	–
`ReminderHistory`	`TaskId`	int → INTEGER	FK → `Tasks.Id`	–
`ReminderHistory`	`RecipientUserId`	int → INTEGER	FK → `Users.Id`	–
`ReminderHistory`	`ReminderType`	`ReminderType` → TEXT(32)	Not Null	FirstReminder, SecondReminder, DueToday, Overdue
`ReminderHistory`	`ScheduledFor`	DateTime → TEXT	Not Null	UTC instant of local midnight of the reminder day
`ReminderHistory`	`Status`	`ReminderStatus` → TEXT(32)	Not Null	The engine writes only `Sent` (Pending/Failed/Skipped are reserved)
`ReminderHistory`	`ProcessedAt`	DateTime? → TEXT	nullable	–
`ReminderHistory`	`DeliveryAttempts`	int → INTEGER	Not Null	Always 0 from the engine
`ReminderHistory`	–	–	**Unique** `(TaskId, RecipientUserId, ReminderType, ScheduledFor)`	Duplicate prevention

Relationships

- `Notifications.UserId → Users.Id` (FK, Cascade)
- `Notifications.TaskId → Tasks.Id` (FK, Cascade)

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- `ReminderHistory.TaskId → Tasks.Id` (FK, Restrict)
- `ReminderHistory.RecipientUserId → Users.Id` (FK, Restrict)
- Referenced by Mail: `MailMessages.ReminderHistoryId → ReminderHistory.Id` (FK, Restrict, unique filtered index) – see `05.2-Mail.md`.
- Writes (reminder mails, owned by Mail): `MailMessages`, `MailRecipients`, `MailboxItems`.
- Reads: `Tasks` (`Status`, `Deadline`, `FirstReminderDays`, `SecondReminderDays`, `OverdueReminderFrequency`, `CreatedByUserId`, `IsDeleted`) and `TaskRecipients` (owned by the Monitoring module). `ApplicationSettings` keys: `Reminder.*`, `Notification.MailEnabled`, `Notification.InAppEnabled`, `Application.Timezone`.

7. Security & Authorization

- **Authentication and authorization:** as described in the parent document (`[Authorize]`, `perm:<key>` policies, `DeniedActionAuditHandler`).
- **Data scope:** always **Own**. Notification queries filter on `UserId == caller`. A notification belonging to someone else returns **404, never 403**.
- **Reminder engine:** runs in-process with no user context and no permission. Its runtime switches are edited in the Settings module under `settings.reminders.manage`.
- **Frontend:** `permissionGuard` on `/notifications`. The menu entry and the header bell are built from `has(Permissions.NotificationsView)`. The Notification Center hides **Mark all as read** with `canManage()`, computed from `AuthService.hasPermission`.

Permission Key	Endpoint(s)	UI Element Controlled
`notifications.view`	`GET /api/v1/notifications`, `GET /api/v1/notifications/unread-count`	Menu "Notifications", route `/notifications`, header bell and unread dot
`notifications.manage`	`POST /api/v1/notifications/{id}/read`, `POST /api/v1/notifications/read-all`	"Mark all as read" button (gated). Per-row "Mark as read" is **not** gated.

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
`page` below 1	`NotificationService`	–	Clamped to 1 (no error)
`pageSize` below 1 or above 100	`NotificationService`	–	Replaced by 20 (no error)
Notification not owned or not found	`NotificationService` / `NotificationsController`	404	– (empty `NotFound()`)
Reminder pass throws	`ReminderSchedulerService.RunPassAsync`	–	Logged as "Reminder pass failed."; the next tick retries

The shared 401 / 403 / 409 / 500 rules are in the parent document §8.

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9. Integrations / Side Effects

- **Reminder engine:** writes `DeadlineReminder` or `Overdue` for each active recipient (the task's recipients plus the creator), once per `(task, recipient, type, day)`.
- **Reminder mails (into Mail):** `IsSystem = true` mails linked to `ReminderHistoryId` are written when the `Notification.MailEnabled` setting is on. That setting is the "Also deliver reminders to Mail" toggle on the Reminder Settings screen.
- **Other modules,** through `INotificationPublisher`:
  - `TaskAssigned` on task create
  - `Reassigned` on reassignment
  - `TaskComment` on comment add, edit or delete
  - `System` on worked-hours change, admin MFA reset and password recovery
- **Audit:** the reminder history (`ReminderHistory`) is the audit record for reminders.
- **AI Assistant:** the help article `notifications.md` in `Services/Assistant/Knowledge/` describes this child module.

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Reminders:SchedulerEnabled`	Turns the background reminder scheduler on or off	`true`
`Reminders:TickSeconds`	Seconds between reminder passes (minimum 1)	`60`
`Mail:AppBaseUrl`	Base web address for the "Open the task:" link in reminder mails	Not set in `appsettings.json` (the link is omitted)

Runtime settings held in the `ApplicationSettings` table, which are edited in the Settings module:

Key	Purpose
`Reminder.Enabled`	Turns the reminder engine on or off
`Reminder.DeadlineDayEnabled`	Sends the due-today reminder
`Reminder.DefaultFirstDays`, `Reminder.DefaultSecondDays`, `Reminder.DefaultOverdueFrequency`	Defaults for new tasks
`Notification.InAppEnabled`	In-app notifications for published events

The shared keys (`Application:Timezone`, `Application.Timezone`, `Notification.MailEnabled`, `Storage:Root`, `ConnectionStrings:Default`) are in the parent document §10.

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11. Testing

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Communication/Notification/Notifications/NotificationServiceTests.cs`	List, paging, mark read, mark all
Unit	`tests/TaskFlow.UnitTests/Communication/Notification/Reminders/ReminderCalculatorTests.cs`	Due-today, first/second, overdue daily/weekly, DST slots
Unit	`tests/TaskFlow.UnitTests/Communication/Notification/Reminders/ReminderServiceTests.cs`	Generation, recipients, mail delivery
Unit	`tests/TaskFlow.UnitTests/Communication/Notification/Reminders/ReminderIdempotencyTests.cs`	Duplicate prevention on repeated passes
Unit	`tests/TaskFlow.UnitTests/Communication/Notification/Reminders/ReminderSchedulerServiceTests.cs`	Scheduler pass and error handling
Unit	`tests/TaskFlow.UnitTests/Monitoring/TaskManagement/Tasks/TaskWorkedHoursNotificationTests.cs`	Worked-hours notifications (cross-module)
Integration	`tests/TaskFlow.IntegrationTests/Communication/Notification/NotificationsEndpointsTests.cs`	Notification endpoints + authorization
Integration	`tests/TaskFlow.IntegrationTests/Settings/ReminderSettings/SettingsEndpointsTests.cs`	Reminder settings that drive the engine

> ⚠️ To be confirmed: the "Covers" column is inferred from file names only, because the test sources were not available. No test file for `NotificationPublisher` appears in `tests-filelist.json`. The cross-module comment test is listed in the parent document §11.

12. Known Limitations / Gaps found

- Notifications are never deleted or archived, and the read time is not stored (no `ReadAt`).
- `ReminderHistory.Status` is always `Sent` and `DeliveryAttempts` is always 0. Retry and failure tracking is not implemented.
- The reminder engine does not push notifications live. The UI picks up new counts on navigation and every 15 seconds.
- `Mail:AppBaseUrl` is not set in `appsettings.json`, so reminder mails carry no task link by default (the reader still shows "Open the task" via `TaskId`).

Gaps found

1. **Reminder notifications bypass the in-app toggle.** `ReminderService.TryDeliverAsync` adds `Notification` rows directly through `INotificationRepository`, not through `NotificationPublisher`. Reminder notifications are therefore still written when `Notification.InAppEnabled` is `false`.
2. **Frontend misses the `TaskComment` type.** `notification.model.ts` `NotificationType` / `NOTIFICATION_TYPES` and `NOTIFICATION_TYPE_LABELS` list only TaskAssigned, Reassigned, DeadlineReminder, Overdue and System. The backend enum also has `TaskComment`, so comment notifications render an empty type chip and cannot be selected in the Type filter.

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3. **Per-row "Mark as read" is not gated.** It is shown to every user with `notifications.view`, but the endpoint requires `notifications.manage` (403). This matches permission-gaps.md, and the code is unchanged.
4. **TDD §8.5 and code differ on mark read.** The TDD lists `PATCH /api/v1/notifications/{id}/read`, but the code uses `POST`. The TDD marks the notification endpoints "Authenticated/Owner", but the code requires the `notifications.view` and `notifications.manage` keys.
5. **The TDD `notifications` table is out of date.** The TDD shows `DedupKey` and `ReadAt` columns and types Reminder/DueToday/Comment. The code has neither column, uses the types `DeadlineReminder` / `TaskComment`, and has no `DueToday` notification type (due-today reminders are written as `DeadlineReminder`).
6. **The TDD `reminder_history` types are out of date.** The TDD names TwoDaysBefore/OneDayBefore. The code uses `FirstReminder` / `SecondReminder` with per-task intervals (`FirstReminderDays`, `SecondReminderDays`) and a Daily/Weekly overdue frequency.
7. **TDD §11.5 recalculation is not implemented as written.** The TDD describes marking `Pending` rows `Skipped`. The code writes no Pending rows. `ReminderCalculator` reads the task's current status and deadline on every pass, and `TryRecordAsync` re-checks the deadline, which gives the same effect.
8. **Reminder recipients differ from TDD and BR-5.** The TDD (§11.5) and BR-5 say `task_recipients` only. The code also adds the task **creator**.
9. **BRD Screen 12 says "Access: All registered users".** In code, access depends on configurable keys, so a custom role without `notifications.view` has no Notification Center. The Administrator, Manager and Employee defaults do match the BRD.
10. **Help knowledge claims status-change notifications.** `notifications.md` says users are notified about "status changes", but no `PublishAsync` call exists for status changes. The help also says "Click a notification to open the task", whereas only the task-code link opens it.

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